1) $a = a_1 \; a_2 \; a_3 \; \cdots \; a_n$
$b = b_1 \; b_2 \; b_3 \; \cdots \; b_n$ ← two n-digit numbers

$$a \times b = \sum_{i=1}^{n} \sum_{j=1}^{n} a_i b_j$$

Number of multiplications:

$a_1$ is multiplied with n numbers

so is $a_2, a_3 \cdots$ etc.

So $n^2$ multiplications are carried out

Then we have $n^2$ numbers to add together in $n^2 - 1$ operations.

So total operations $= 2n^2 - 1$

So the time complexity has order $n^2$

2) $f \in C^{\infty}([a,b], \mathbb{R})$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \qquad x_n \to \alpha$$

$$f(\alpha) = 0$$

Let the error $e_n := x_n - \alpha$

a) The taylor expansion of $g$ around $\alpha$ is

$$f(x_n) = f'(\alpha) e_n + \frac{1}{2} f''(\alpha) e_n^2 + O(e_n^3)$$

$$f'(x_n) = f'(\alpha) + f''(\alpha) e_n + O(e_n^2)$$

$$\frac{f(x_n)}{f'(x_n)} = \frac{f'(\alpha) e_n + \frac{1}{2} f''(\alpha) e_n^2 + O(e_n^3)}{f'(\alpha) + f''(\alpha) e_n + O(e_n^2)}$$

$$= e_n \left( \frac{1 + \frac{1}{2} \frac{f''(\alpha)}{f'(\alpha)} e_n + O(e_n^2)}{1 + \frac{f''(\alpha)}{f'(\alpha)} e_n + O(e_n^2)} \right)$$

$$\frac{1+a}{1+b} = 1 + (a-b) + O(e_n^2)$$

so $$\frac{f(x_n)}{f'(x_n)} = e_n \left( 1 + \frac{1}{2} \frac{f''(\alpha)}{f'(\alpha)} e_n + O(e_n^2) - \frac{f''(\alpha)}{f'(\alpha)} e_n - O(e_n^2) \right)$$

$$= e_n \left( 1 - \frac{1}{2} \frac{f''(\alpha)}{f'(\alpha)} e_n + O(e_n^2) \right)$$

$$= e_n - \frac{1}{2} \frac{f''(\alpha)}{f'(\alpha)} e_n^2 + O(e_n^3)$$

$$e_{n+1} = e_n - \left(e_n - \frac{1}{2}\frac{f''(\alpha)}{f'(\alpha)} e_n^2 + O(e_n^3)\right)$$

$$= \frac{1}{2}\frac{f''(\alpha)}{f'(\alpha)} e_n^2 + O(e_n^3)$$

So quadratic convergence ✓

b) If $f'(\alpha) = 0$ and $f''(\alpha) \neq 0$

The Taylor expansion becomes

$$f(x_n) = \frac{1}{2} f''(\alpha) e_n^2 + O(e_n^3)$$

$$f'(x_n) = f''(\alpha) e_n + O(e_n^2)$$

Then $$\frac{f(x_n)}{f'(x_n)} = \frac{\frac{1}{2} f''(\alpha) e_n^2 + O(e_n^3)}{f''(\alpha) e_n + O(e_n^2)}$$

$$= \frac{1}{2} e_n + O(e_n^2)$$

$$e_{n+1} = e_n - \frac{1}{2} e_n + O(e_n^2)$$

$$= \frac{1}{2} e_n + O(e_n^2)$$

$e_{n+1} \approx \frac{1}{2} e_n$ so linear convergence with the error halving each time.

c) If $f'(\alpha) = 0$, $f''(\alpha) = 0$, $f'''(\alpha) \neq 0$:

The taylor expansion becomes

$$f(x_n) = \frac{1}{6} f'''(\alpha) e_n^3 + O(e_n^4)$$

$$f'(x_n) = \frac{1}{2} f'''(\alpha) e_n^2 + O(e_n^3)$$

So $$\frac{f(x_n)}{f'(x_n)} = \frac{\frac{1}{6} f'''(\alpha) e_n^3 + O(e_n^4)}{\frac{1}{2} f'''(\alpha) e_n^2 + O(e_n^3)}$$

$$= \frac{1}{3} e_n + O(e_n^2)$$

So $$e_{n+1} = e_n - \frac{1}{3} e_n - O(e_n^2)$$

$$= \frac{2}{3} e_n - O(e_n^2)$$

$e_{n+1} \approx \frac{1}{3} e_n$ so linear convergence but slower than if $f''(\alpha) \neq 0$

d) If $f'(\alpha) \neq 0$, $f''(\alpha) = 0$

Then the taylor expansion becomes

$$f(x_n) = f'(\alpha) e_n + \frac{1}{6} f'''(\alpha) e_n^3 + O(e_n^4)$$
$$f'(x_n) = f'(\alpha) + f'''(\alpha) e_n^2 + O(e_n^3)$$

$$\frac{f(x_n)}{f'(x_n)} = \frac{f'(\alpha) e_n + \frac{1}{6} f'''(\alpha) e_n^3 + O(e_n^4)}{f'(\alpha) + f'''(\alpha) e_n^2 + O(e_n^3)}$$

$$= e_n \left( \frac{1 + \frac{1}{6} \frac{f'''(\alpha)}{f'(\alpha)} e_n^2 + O(e_n^3)}{1 + \frac{f'''(\alpha) e_n^2}{f'(\alpha)} + O(e_n^3)} \right)$$

$$= e_n \left( 1 + \frac{1}{6} \frac{f'''(\alpha)}{f'(\alpha)} e_n^2 + O(e_n^3) - \frac{f'''(\alpha) e_n^2}{f'(\alpha)} - O(e_n^3) \right)$$

$$= e_n - \frac{5}{6} \frac{f'''(\alpha)}{f'(\alpha)} e_n^3 + O(e_n^4)$$

$$e_{n+1} = e_n - \left( e_n - \frac{5}{6} \frac{f'''(\alpha)}{f'(\alpha)} e_n^3 + O(e_n^4) \right)$$

$$= \frac{5}{6} \frac{f'''(\alpha)}{f'(\alpha)} e_n^3 + O(e_n^4)$$

So $e_{n+1} \approx \frac{5}{6} e_n^3$ ← cubic convergence